Structurally Constrained PINNs for Barrier Option Pricing Under a High-Precision Implicit Finite-Difference Benchmark

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Abstract

Continuously monitored barrier options are a local-precision stress test rather than a generic smooth pricing task. The absorbing barrier creates a narrow region in which both the option value and local sensitivities vary sharply. We study down-and-out European calls in a one-dimensional Black–Scholes setting and evaluate neural surrogates against a deliberately strong classical benchmark. The benchmark uses a transformed log-domain formulation, exact barrier alignment, and Rannacher-smoothed Crank–Nicolson time stepping. The proposed neural framework combines transformed coordinates, a barrier-preserving hard-constrained ansatz, barrier-aware adaptive collocation, and hybrid optimization. Evaluation is protocol-based: prices, Delta, Gamma, barrier consistency, positivity, residual diagnostics, and deployment economics are assessed jointly rather than through pricing error alone. The benchmark FDM remains strongest for absolute local accuracy and Greek-sensitive tasks. Among learned models, the barrier-aware PINN is strongest in near-barrier Gamma control and boundary consistency, while supervised surrogates become economically attractive only under sufficiently large repeated-query workloads and conditional error tolerance. The main conclusion is therefore not neural replacement, but structured solver selection in which accuracy determines admissibility and amortization determines economic preference.

Keywords: barrier option pricing; physics-informed neural networks; implicit finite differences; validation protocol; deployment economics

1. Introduction

Barrier options are numerically difficult for a reason that is more structural than exotic-label driven. A continuously monitored barrier creates an absorbing constraint over the full life of the contract, not merely at maturity. As a result, the pricing surface develops a narrow layer near the barrier in which both the value function and local sensitivities vary sharply. In practical terms, the key challenge is therefore not broad average approximation across the full state space, but stable local control in the most fragile regions, especially near the barrier and around the strike (Broadie et al., 1997; Reiner & Rubinstein, 1991).

This distinction matters for both numerical analysis and financial use. A model can exhibit acceptable global price error while still producing unstable Delta or Gamma precisely where hedging and risk reporting are most sensitive. We therefore treat barrier pricing as a local-precision stress test for both classical solvers and neural surrogates.

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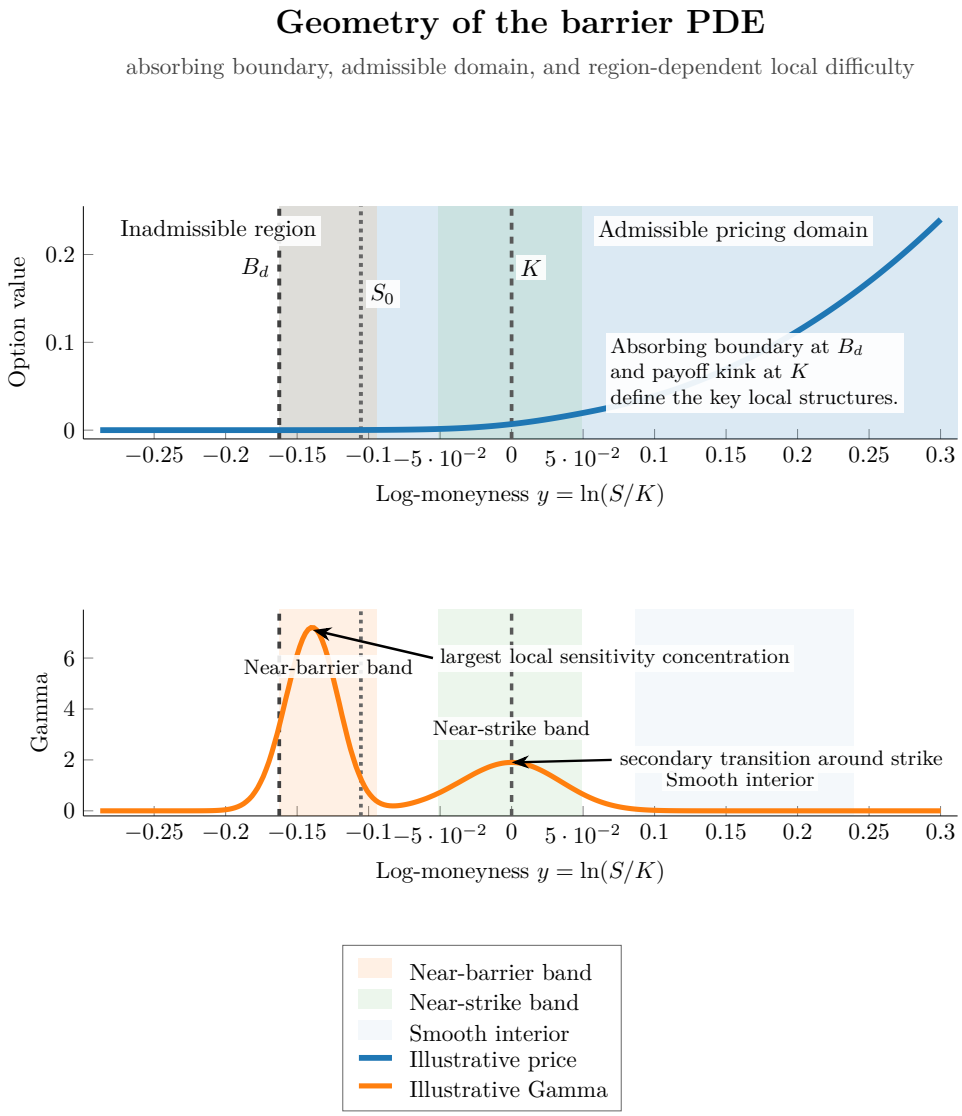

Figure 1: Geometry of the barrier PDE. The barrier layer and the near-strike region create localized numerical difficulty, while the smooth interior is substantially easier.

Classical implicit finite differences remain the natural benchmark in this setting. In one-dimensional Black–Scholes problems, carefully implemented Crank–Nicolson schemes with barrier alignment and Rannacher smoothing remain highly accurate and professionally defensible (Duffy, 2006; Tavella & Randall, 2000; Zvan et al., 2000). Their weakness is repeated-query cost rather than numerical quality. Neural surrogates are attractive because they can shift cost from repeated solves to offline training, but barrier pricing is exactly the type of problem in which runtime claims should not be accepted without a strong comparator and a validation protocol that goes beyond pricing error alone.

The methodological gap addressed here is therefore specific. Existing neural-PDE and neural-pricing work often demonstrates promising average approximation or runtime gains, but much less often studies barrier-specific failure modes under a strong benchmark while jointly validating prices, Greeks, boundary consistency, residual diagnostics, and deployment economics. More recent finance-facing neural-pricing studies have also begun to examine surrogate deployment in stochastic-volatility and large-scale option en-

vironments (Chen et al., 2026; Hainaut & Casas, 2024). This paper addresses that gap in a controlled but demanding one-dimensional setting with analytical truth available.

The paper makes four contributions. First, it constructs a strong implicit finite-difference benchmark rather than a weak classical baseline. Second, it identifies why naive barrier PINNs fail and proposes a barrier-aware surrogate combining transformed coordinates, a hard-constrained ansatz, BAAC, and hybrid optimization. Third, it evaluates surrogates under a protocol in which admissibility is a joint decision across price, Gamma, boundary, positivity, and residual diagnostics. Fourth, it quantifies offline–online amortization and shows when a learned surrogate becomes economically interesting relative to the benchmark.

Operationally, the paper compares three computational logics. The first is classical implicit FDM, which is sequential but highly reliable. The second is a physics-informed surrogate, which tries to preserve barrier structure while shifting cost offline. The third is label-driven supervised approximation, which can become highly efficient once a trusted benchmark exists. The point of the paper is not to crown one paradigm unconditionally, but to show how barrier geometry changes what counts as a credible comparison among them.

The analysis is organized around four questions. Why do naive barrier PINNs fail in the hardest regimes? Which structural constraints are necessary to stabilize them? How should a barrier surrogate be validated before it is accepted? And under what workload regime does offline training begin to alter the preferred solver? These questions provide the organizing logic for the sections that follow.

2. Problem Setup and Benchmark Institution

2.1. Contract, PDE, and Analytical Reference

We study a continuously monitored down-and-out European call with strike $K > 0$, maturity $T > 0$, lower barrier $B_d > 0$, and zero rebate. Under the risk-neutral measure,

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}} \tag{1}$$

and the surviving contract value $V(S,t)$ satisfies

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r-q)S\frac{\partial V}{\partial S} - rV = 0 \qquad S \in (B_d, \infty) \quad t \in [0,T) \tag{2}$$

The terminal and boundary conditions are

$$V(S,T) = \max(S - K, 0) \qquad S > B_d \tag{3}$$

$$V(B_d, t) = 0 \qquad t \in [0,T] \tag{4}$$

$$V(S,t) \sim Se^{-q(T-t)} - Ke^{-r(T-t)} \quad \text{as } S \to \infty \tag{5}$$

We use the transformed coordinates

$$x = \frac{S}{K} \qquad y = \ln\left(\frac{S}{K}\right) \qquad \beta = \frac{B_d}{K} \tag{6}$$

because the logarithmic coordinate removes the raw quadratic diffusion scaling of the original asset-price operator and stabilizes both classical and neural treatment.

Analytical truth is available through the Reiner–Rubinstein closed form for down-and-out calls, which makes this problem unusually suitable for disciplined benchmarking.

That analytical layer is used for out-of-sample price evaluation and for numerical Greek comparisons.

This analytical reference plays two roles. First, it provides price truth without introducing benchmark noise into the primary comparison. Second, it allows the study to distinguish benchmark verification from surrogate evaluation. The FDM solver is therefore not used as an unquestioned oracle; it is itself checked against analytical truth before it is used as the classical benchmark for the broader solver family.

2.2. Why the Problem Is Numerically Hard

The barrier problem is difficult because error is not uniformly distributed across the domain. The near-barrier layer is fragile because survival value changes rapidly with small perturbations in state. The near-strike band remains sensitive because of payoff non-smoothness. The smooth interior is comparatively benign. A credible solver must therefore be judged regionally, not just by a single global average.

This geometry is exactly why barrier pricing is informative for neural surrogates. A generic function approximator can do well in broad interior regions and still fail the task that matters most. In the present setting, this is visible most clearly in near-barrier Gamma behavior, barrier residuals, and upper-tail pricing error.

There is also a practical interpretation behind this geometry. Near the barrier, small local mispricing translates directly into survival-probability distortion. Around the strike, local curvature matters for hedge stability. In the smooth interior, by contrast, approximation is much more forgiving. This asymmetry is why the paper repeatedly privileges upper-tail and regional diagnostics over a single mean statistic.

2.3. High-Precision Implicit FDM Benchmark

The classical benchmark is designed to be deliberately strong rather than merely serviceable. The implementation combines four choices:

1. a log-domain transformed formulation for operator regularity,
2. exact barrier alignment so that the absorbing boundary is enforced directly at a grid node,
3. Rannacher-smoothed Crank–Nicolson time stepping to stabilize the payoff kink,
4. sparse pre-factorization for efficient repeated backward propagation.

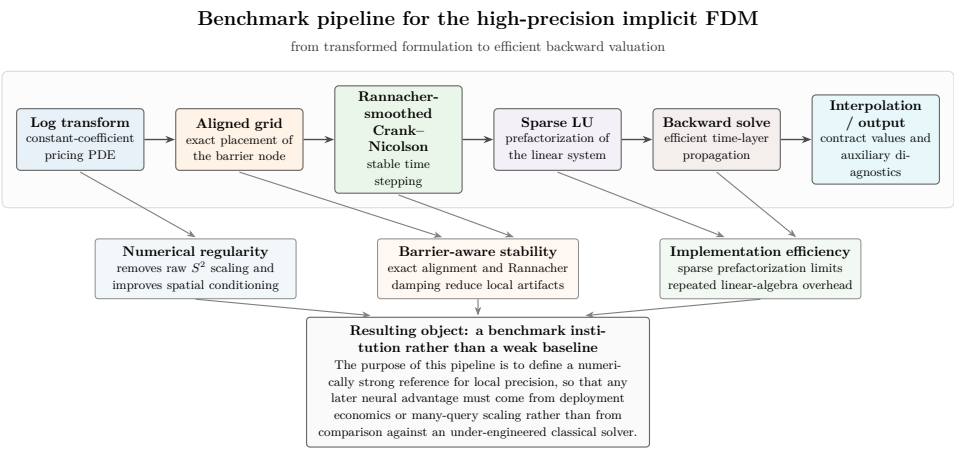

Figure 2: Benchmark pipeline for the high-precision implicit finite-difference solver.

This design matters methodologically as much as numerically. If the classical comparator is weak, favorable neural conclusions are not informative. If the classical comparator is strong, any surrogate claim must survive a much more meaningful standard.

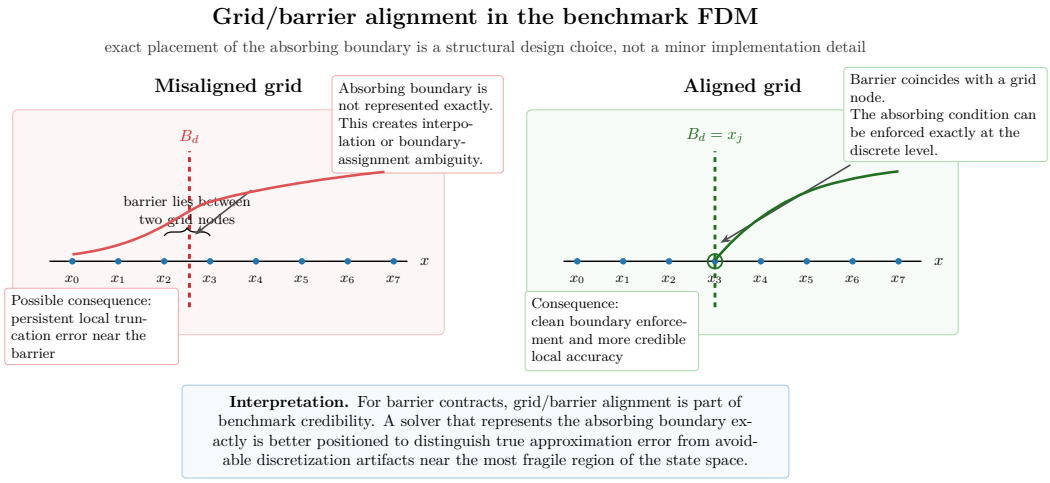

Figure 3: Grid/barrier alignment in the benchmark solver. Exact alignment removes an avoidable source of ambiguity near the absorbing boundary.

Exact barrier alignment deserves special emphasis. In a misaligned grid, some of what appears to be model error is really preventable discretization ambiguity caused by how the absorbing condition is represented. Aligning the barrier with a grid node removes that ambiguity and makes the benchmark cleaner as an evidentiary reference.

2.4. Benchmark Verification

The retained benchmark configuration is $(N_x, N_t) = (800, 2000)$. It lies on a favorable accuracy–runtime frontier: the benchmark is already highly accurate while avoiding unnecessary solve cost. The broader convergence study shows median relative pricing error improving from approximately $8.06 \times 10^{-3}\%$ at $(200, 500)$ to $2.33 \times 10^{-4}\%$ at $(1600, 4000)$, with 95th percentile relative error declining from about 0.119% to 0.0139%. At the retained configuration, the 95th percentile relative pricing error is about $2.06 \times 10^{-2}\%$, which is already sufficiently small for the benchmark role.

This retained configuration should not be interpreted as the finest grid attainable, but as the benchmark point that is strong enough for the comparison while remaining operationally reasonable. That distinction is important later in the runtime section. The benchmark is not optimized for the cheapest possible solve, nor is it chosen to be unrealistically expensive; it is chosen to be a credible numerical institution against which surrogate claims remain meaningful.

Table 1: Compressed benchmark verification summary for the retained FDM benchmark.

Quantity	Retained conclusion
Spatial-temporal scheme	Log-domain implicit Crank–Nicolson with Rannacher smoothing
Barrier treatment	Exact grid alignment at the absorbing boundary
Retained configuration	$(N_x, N_t) = (800, 2000)$
Median relative pricing error	On the order of $10^{-3}\%$
95th percentile relative pricing error	Approximately $2.06 \times 10^{-2}\%$
Greek behavior	Small Delta and Gamma errors under refinement
Role in the paper	Strong benchmark institution for local precision

3. Structurally Constrained Barrier-Aware Neural Surrogate

3.1. Transformed Parametric Formulation

The surrogate is constructed in transformed coordinates and parameterized across contract families. Rather than training a separate network for each contract, the retained design follows a one-network-many-contract view in which the network takes both state variables and contract parameters as input. This is important for deployment relevance: the surrogate should approximate a family-level mapping, not just a single fixed-instance PDE solution.

We work with transformed coordinates and a non-dimensionalized price representation in order to reduce operator-scaling imbalance. The ablation evidence below shows that this is necessary rather than cosmetic: the raw S-space formulation fails decisively, while transformed coordinates move the model from failure toward partial numerical credibility.

The parametric view also changes how the neural solver should be judged. A fixed-instance PINN may still be acceptable as a methodological test, but it is not naturally aligned with repeated-query deployment over contract panels. A parameterized surrogate is therefore more appropriate for the workload interpretation developed later, even in the one-dimensional setting studied here.

3.2. Barrier-Preserving Neural Ansatz

The central structural design choice is a hard-constrained barrier ansatz. Instead of asking the optimizer to discover the absorbing boundary through a soft penalty alone, the surrogate is written as

$$\widehat{V}_\theta(x, \tau, \mu) = \psi_{\text{bar}}(x; \mu)\, A_\theta(x, \tau, \mu) \tag{7}$$

where the barrier factor ψ_{bar} vanishes at the barrier and the neural amplitude A_θ is trainable. This guarantees the knockout condition by construction and turns a fragile learning target into a structural property of the hypothesis class.

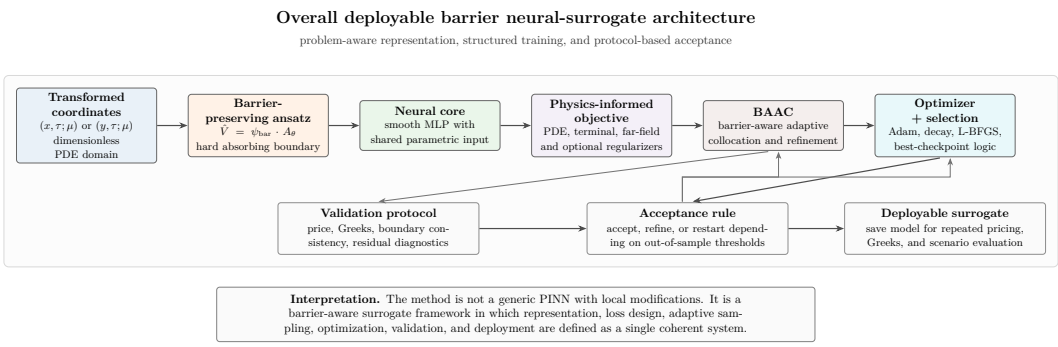

Figure 4: Overall deployable barrier neural-surrogate architecture. The method is designed as an integrated system rather than a generic PINN with local patches.

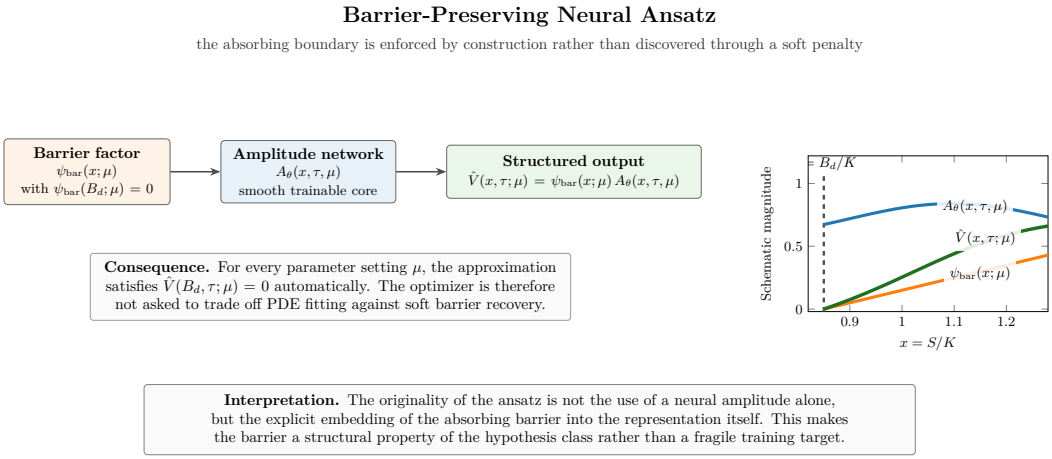

Figure 5: Barrier-preserving neural ansatz. The barrier is embedded into the representation rather than recovered by a soft penalty.

This is the single most important structural distinction between the retained barrier-aware PINN and a naive penalty-based PINN. In the barrier problem, soft enforcement alone is too unstable because optimization must simultaneously fit the interior PDE and a sharp absorbing boundary.

In effect, the ansatz changes the learning problem. Instead of solving for a surface that merely should satisfy the barrier, the model solves for a surface inside a class that already respects it. This reduces optimization burden and improves interpretability: remaining errors can be attributed to interior approximation and sampling weakness rather than to basic failure to honor the contract structure.

3.3. Physics-Informed Objective, BAAC, and Hybrid Training

The loss function combines the interior PDE residual with terminal, far-field, and auxiliary structural terms. In the compressed manuscript, the important point is not the exact bookkeeping of every coefficient, but the way training is made barrier-aware. The retained method uses:

1. transformed coordinates to improve conditioning,
2. a hard-constrained ansatz to enforce the barrier exactly,

3. smooth activations and region-aware features,
4. barrier-aware adaptive collocation (BAAC),
5. hybrid Adam/L-BFGS optimization with validation-based checkpointing.

BAAC is critical because the difficult region is spatially localized. Sampling combines broad interior coverage with explicit near-barrier and near-strike quotas, and then refines collocation density where residuals remain large.

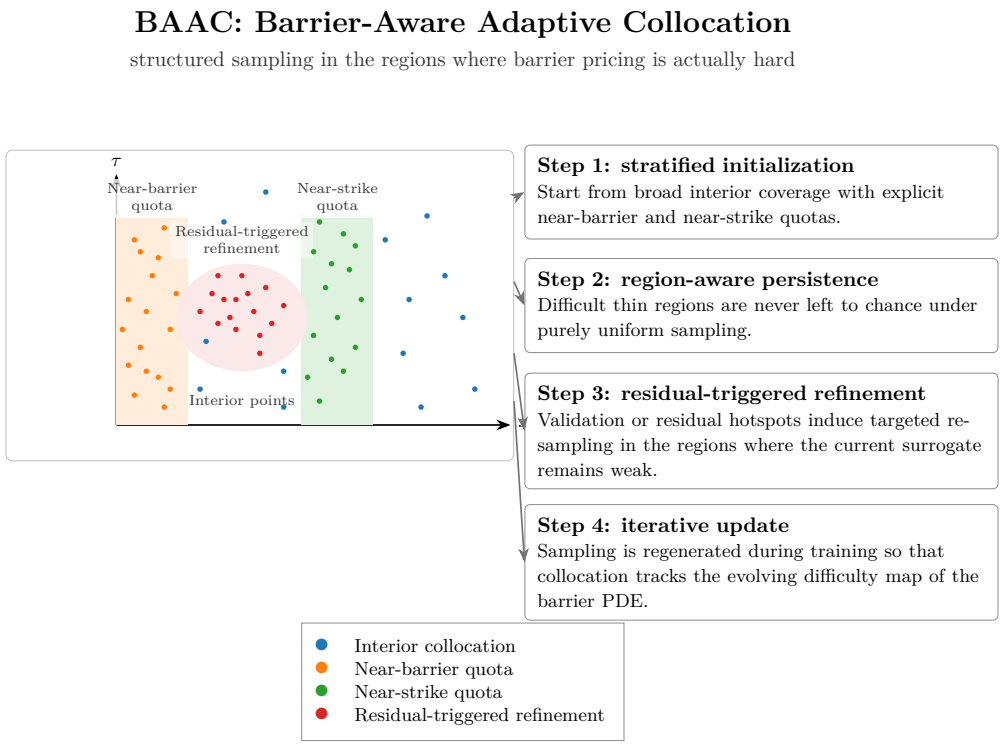

Figure 6: Barrier-aware adaptive collocation (BAAC).

3.4. Baseline Family

The comparison family includes four models:

1. the high-precision FDM benchmark,
2. the barrier-aware PINN,
3. a structured supervised surrogate trained on benchmark labels,
4. a stabilized differential surrogate using price-plus-Delta supervision.

This family is deliberately heterogeneous. FDM provides the local-precision reference, the PINN tests whether barrier structure can be encoded credibly in a physics-informed surrogate, and the supervised models test whether label-driven surrogates remain attractive once hard structure is imposed and repeated-query economics are considered.

Table 2: Baseline family and comparison roles.

Model	Training signal	Role in the comparison
FDM benchmark	None; direct numerical solve	Classical reference for local precision and Greek stability
Barrier-aware PINN	PDE residual plus structural constraints	Tests whether barrier structure can be encoded without dense labels
Structured supervised surrogate	Benchmark labels	Tests smooth-surface approximation once hard boundary structure is imposed
Differential surrogate	Price-plus-Delta labels	Tests whether derivative-aware supervision improves local control materially

4. Experimental Design and Validation Protocol

4.1. Data Split, Scenario Family, and Regional Diagnostics

The evaluation design is protocol-based. Training, validation, test, and stress panels are separated so that acceptance is not based on the same points used to fit the model. The scenario family is organized around four groups: core scenarios, near-barrier stress, short-maturity/low-volatility stress, and a wider random panel. This keeps the experimental design aligned with the actual geometry of the problem rather than with a few ad hoc examples.

Validation is regionalized into near-barrier, near-strike, smooth interior, and far-field zones. This is essential. A single global average can hide severe failure in the barrier layer.

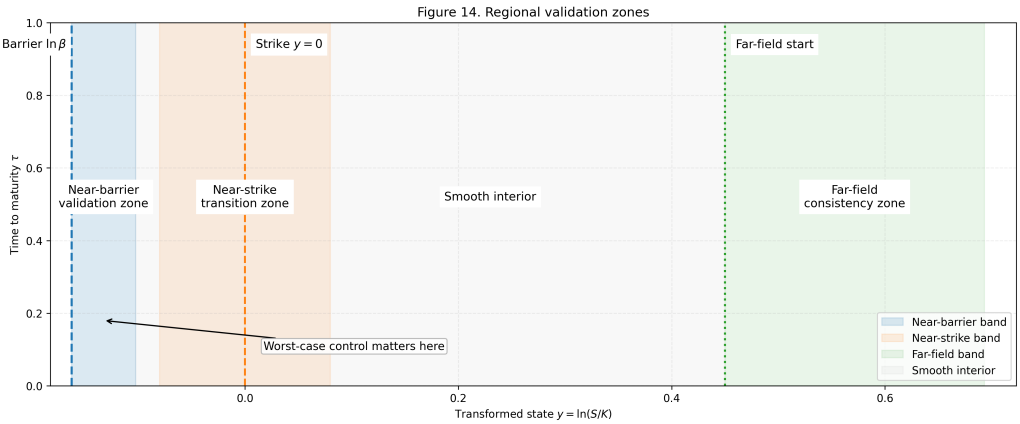

Figure 7: Regional validation zones. The protocol distinguishes the barrier layer from the near-strike band and the smoother interior.

The split design also has an evidentiary role. Training establishes whether the model can fit the retained workflow. Validation governs checkpoint selection and acceptance. Test data supports the main reported metrics. Stress panels probe the regions in which the problem geometry suggests that smooth interpolation should not be trusted. This separation is stricter than a conventional single-panel experiment and is important because barrier credibility is easy to overstate when training and evaluation are too closely coupled.

4.2. Metrics and Acceptance Rule

The validation protocol jointly evaluates:

- price relative error,
- Delta and Gamma error,
- barrier residual,
- positivity violation rate,
- PDE residual diagnostics,
- runtime and break-even quantities for deployment interpretation.

The retained acceptance thresholds are:

$$\text{Validation price q95 relative error} \leq 2.00\% \tag{8}$$

$$\text{Near-barrier Gamma q95 absolute error} \leq 0.150 \tag{9}$$

$$\max |V(B_d, t)| \leq 10^{-5} \tag{10}$$

$$\text{Validation PDE residual q95} \leq 10^{-2} \tag{11}$$

$$\text{Positivity violation rate} = 0 \tag{12}$$

Not every metric is used as a hard threshold, but the protocol forces the paper to distinguish acceptable from merely interesting behavior. In particular, it prevents the common mistake of promoting a model based only on mean pricing fit or selected plots.

The protocol also explains why the paper relies heavily on q95 statistics. In a barrier problem, upper-tail behavior in difficult regions is often more informative than a small mean error diluted by easy interior points. The same logic applies to the barrier residual: exact satisfaction is not itself sufficient for a good model, but failure there is immediately disqualifying because it contradicts the contract structure.

Table 3: Compressed validation and experiment summary.

Item	Retained specification
Data design	Train / validation / test / stress separation
Regional zones	Near-barrier, near-strike, smooth interior, far-field
Core pricing threshold	Validation price q95 $\leq 2.00\%$
Core Greek threshold	Near-barrier Gamma q95 ≤ 0.150
Boundary threshold	Barrier residual max $\leq 10^{-5}$
Residual threshold	Validation PDE residual q95 $\leq 10^{-2}$
Admissibility threshold	Positivity violation rate $= 0$
Protocol logic	A model is accepted only if all core criteria pass

Table 4: Scenario families retained in the compressed manuscript.

Family	Purpose	Main diagnostic role
Core scenarios	Representative benchmark panel	Anchor the main price comparison across solver families
Near-barrier stress	Small ρ_d and fragile boundary regimes	Expose localized failure in price and Gamma
Short-maturity / low-volatility stress	Difficult temporal-curvature regimes	Test whether learned models remain stable beyond the core panel
Wide random panel	Broader parameter coverage	Support out-of-sample and upper-tail evaluation

5. Results

5.1. Ablation: From Naive Failure to the Retained Design

The ablation logic is organized around four questions: why the naive PINN fails, why transformed coordinates matter, why hard barrier enforcement is necessary, and why BAAC improves regional control. The main message is cumulative: the retained model is not the product of one isolated trick, but of addressing a sequence of interacting failure modes.

Table 5: Compressed ablation summary preserving the decisive retained variants. Lower is better for all error and violation metrics.

Variant	Median RE (%)	95th RE (%)	Worst-case RE (%)	Delta error	Gamma error	Barrier residual	Positivity violation
Naive PINN	63.337000	203.319000	1332.372000	0.307800	0.584800	3.836200	0.050000
Raw S-space	100.000000	100.000000	100.000000	1.208700	0.018200	8.380000×10^{-18}	0.000000
$y = \ln(S/K)$	3.728000	4.821000	39.275000	0.062100	0.015200	9.670000×10^{-6}	0.000000
Soft BC	80.396000	104.696000	1300.464000	1.177500	0.024900	6.356400	0.262500
Hard barrier only	4.925000	6.470000	25.053000	0.056100	0.015100	1.020000×10^{-5}	0.000000
Full BAAC	3.645000	4.912000	41.397000	0.043100	0.014100	9.680000×10^{-6}	0.000000

The retained summary keeps the variants that determine the design logic: naive failure, coordinate transformation, hard barrier enforcement, and full BAAC.

Three conclusions follow. First, naive barrier PINNs fail systematically: the median price error is over 63%, the 95th percentile exceeds 200%, and boundary behavior is poor. Second, transformed coordinates are necessary. The raw S-space model effectively collapses, whereas the log-domain model reduces median error to about 3.73%. Third, hard barrier enforcement is indispensable. The soft-boundary variant remains numerically and financially weak, while the hard-constrained form restores structural admissibility. BAAC then improves regional balance further, reducing median price error to about 3.65%.

The table is best read as a causal chain. The naive PINN fails because the optimization problem is poorly scaled, the absorbing boundary is treated too softly, and the sampling design under-resolves the most fragile regions. Coordinate transformation addresses the first issue; the hard-constrained ansatz addresses the second; BAAC addresses the third. The point is not that every metric improves monotonically at every step, but that the retained model family emerges naturally from diagnosing the dominant failure channels.

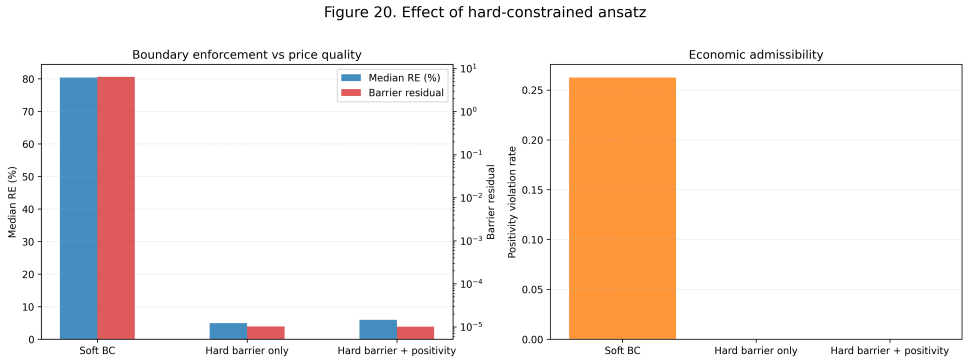

Figure 8: Effect of the hard-constrained ansatz. Hard structural enforcement sharply improves boundary behavior and admissibility.

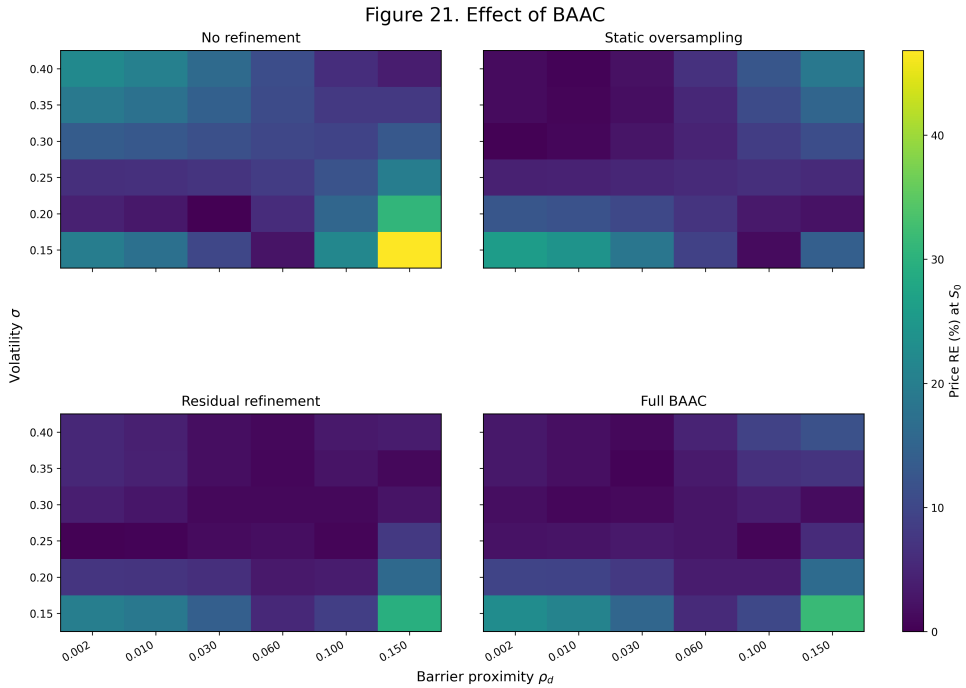

Figure 9: Effect of BAAC. Barrier-aware sampling improves spatial robustness in the most difficult parameter regions.

One nuance deserves mention. The positivity-preserving variant explored in the longer manuscript eliminates financial inadmissibility cleanly, but in the retained run it does not uniformly dominate the simpler hard-barrier-only design in price metrics. This is informative rather than disappointing: positivity control is valuable, but it is not free. It should be understood as a financially motivated structural enhancement rather than as an automatic improvement in every numerical metric.

5.2. Pricing Accuracy and Regional Error Concentration

Table 6 reports the six representative core scenarios retained from the full manuscript. The hierarchy is clear. The FDM benchmark remains extremely accurate across all cases; the learned models are much more heterogeneous.

Table 6: Main pricing comparison against analytical truth for the six representative core scenarios.

σ	ρ_d	B_d	T	Truth	FDM RE (%)	PINN RE (%)	Supervised RE (%)	Differential RE (%)
0.15	0.002	99.8	1	0.484456	0.034695	25.783247	38.301361	37.440460
0.15	0.150	85.0	1	11.570767	0.000244	17.191073	15.595046	10.392541
0.25	0.002	99.8	1	0.343278	0.012653	3.711658	23.354402	24.080633
0.25	0.150	85.0	1	13.563143	0.000118	1.254342	0.785593	0.772599
0.40	0.002	99.8	1	0.278499	0.002244	2.090086	22.278642	27.303123
0.40	0.150	85.0	1	14.813006	0.013634	13.058969	4.157883	0.843983

RE = relative error. The full thesis version reports corresponding absolute errors and runtimes as well; the compressed table keeps the quantities most central to the pricing comparison.

The FDM benchmark remains below 0.04% relative error in every reported scenario. By contrast, the learned models split sharply between smooth and difficult regimes. In moderate interior cases such as $(\sigma, \rho_d) = (0.25, 0.15)$, the supervised and differential surrogates achieve relative errors below 1%, and the PINN remains competitive at about 1.25%. In the hardest near-barrier case $(0.15, 0.002)$, however, the PINN rises to about 25.78% while the supervised and differential surrogates deteriorate further to approximately 38.30% and 37.44%.

The main implication is not that all learned models are uniformly poor. Rather, they fail where the problem is actually hardest. In barrier pricing, that matters more than mean behavior in easy regions.

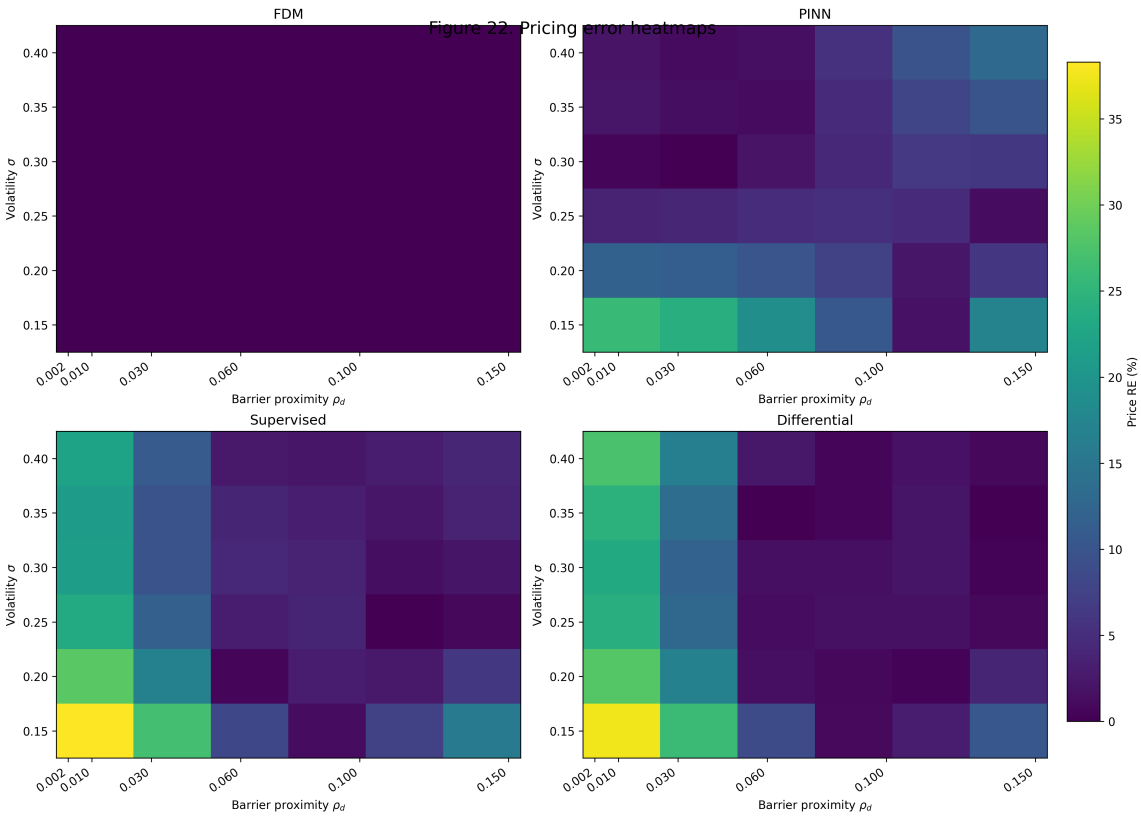

Figure 10: Pricing error heatmaps across the (ρ_d, σ) grid. Learned-model error concentrates in the low-ρ_d barrier layer.

Figure 10 makes this regional structure immediate. The learned models do not merely have higher average error; they have error concentrated in the barrier layer. The barrier-aware PINN deteriorates there, but materially less than the label-driven surrogates.

This asymmetry is exactly why plain leaderboard-style comparison is misleading in the barrier setting. A model can look competitive on a broad interior average while still being the wrong tool for the financially sensitive region. The price comparison is therefore necessary but not sufficient; it needs to be read together with the Greek and boundary diagnostics that follow.

5.3. Greeks, Boundary Consistency, and Residual Diagnostics

Gamma is the most discriminating local diagnostic in this study. Price fit alone can be misleading; barrier credibility is much more tightly connected to whether a model remains stable in near-barrier curvature.

Delta provides an intermediate diagnostic. In the full manuscript, Delta is less unforgiving than Gamma but still more informative than price alone. The same ordering seen in the pricing results reappears there: FDM remains strongest, the barrier-aware PINN is the best learned model in the difficult regions, and the label-driven surrogates are more acceptable in smoother interior zones than in the barrier layer. Gamma then sharpens that ranking further and makes the near-barrier gap impossible to ignore.

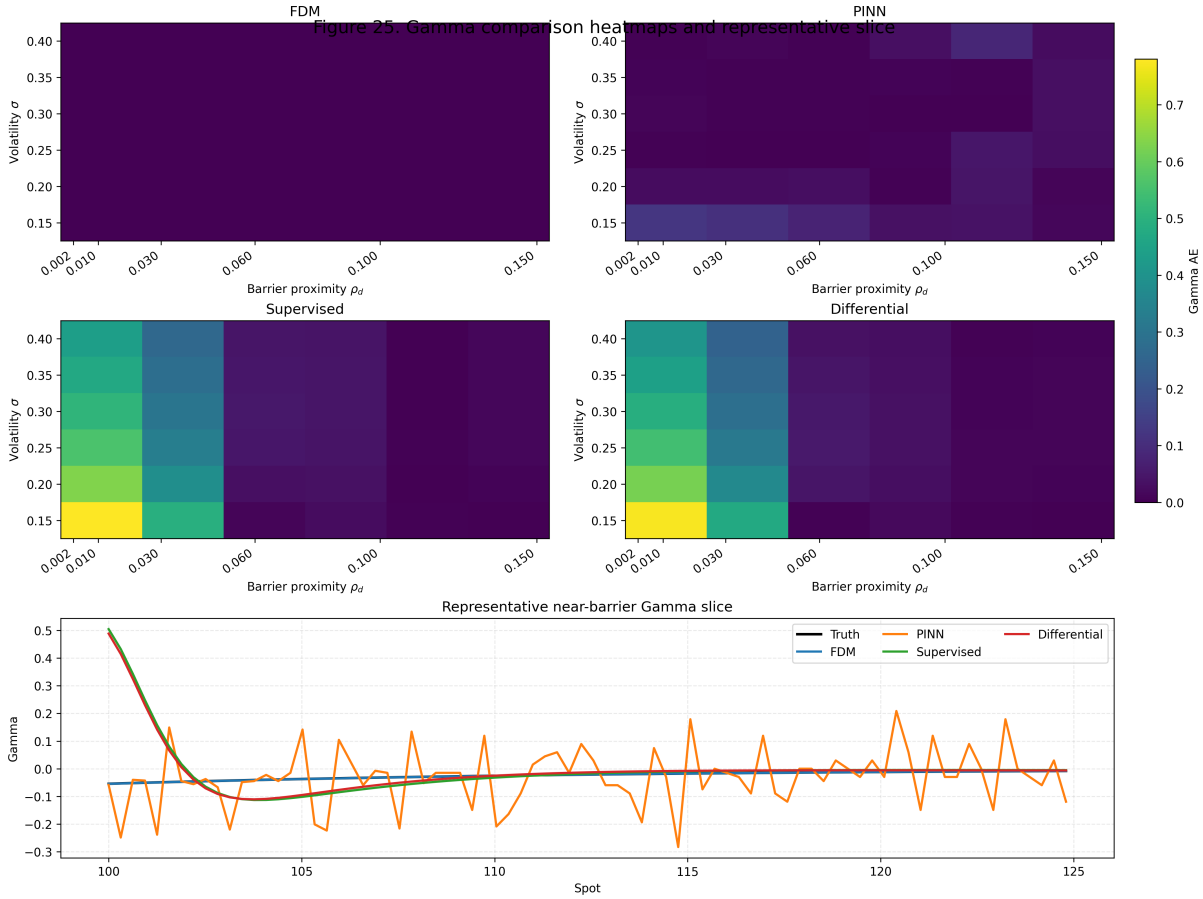

Figure 11: Gamma comparison heatmaps and representative slice. The barrier-aware PINN is the strongest learned model in the near-barrier Gamma diagnostic, although it still trails the FDM benchmark.

The validation scorecard captures this ordering succinctly.

Table 7: Validation scorecard under the retained protocol.

Model	Price q95		Near-barrier Gamma q95		Barrier residual max		Residual q95		Positivity		Overall
	Value	Pass	Value	Pass	Value	Pass	Value	Pass	Rate	Pass	
FDM	0.017924	PASS	0.000723	PASS	0.000000	PASS	0.509095	FAIL	0.000000	PASS	FAIL
Supervised	18.977771	FAIL	0.747259	FAIL	0.000006	PASS	132.133696	FAIL	0.000000	PASS	FAIL
Differential	18.604733	FAIL	0.738783	FAIL	0.000006	PASS	133.814411	FAIL	0.000000	PASS	FAIL
PINN	21.322392	FAIL	0.106243	PASS	0.000009	PASS	9.957489	FAIL	0.000000	PASS	FAIL

The residual criterion is known to be over-severe in the near-barrier layer, but it is retained because it still provides useful diagnostic ordering outside the most singular region.

Several conclusions are important. First, no learned model passes the full protocol. Second, the FDM benchmark remains the only model that simultaneously passes pricing accuracy, near-barrier Gamma, and boundary consistency. Third, among learned models, the barrier-aware PINN is clearly the strongest in the barrier-specific diagnostics: it passes the near-barrier Gamma threshold with q95 error approximately 0.106, passes the barrier residual threshold, and preserves positivity. The supervised and differential surrogates also satisfy the barrier test after hard structure is imposed, but they fail sharply on price q95 and Gamma q95. The FDM barrier residual is on the order of 4.84×10^{-14}, while the learned models that incorporate hard barrier structure remain near the prescribed admissibility scale. This confirms that the ansatz is doing real work, but also that boundary pass alone does not imply full local credibility.

Residual diagnostics require slightly more caution. The retained residual estimator is intentionally conservative near the barrier and therefore too severe to serve as a final certification device. Even so, it remains informative in two ways. First, it warns against overclaiming when the operator fit remains fragile. Second, it still preserves useful ordering outside the most singular region. For that reason the residual criterion remains in the protocol, but its interpretation is explicitly limited.

6. Deployment Economics, Limitations, and Conclusion

6.1. Runtime and Break-Even Analysis

The runtime comparison distinguishes one-off evaluation from repeated-query deployment. Under the retained host and implementation, the representative latencies are

$$c_{\text{fdm}} \approx 0.00345 \text{ s} \tag{13}$$

$$c_{\text{inf,pinn}} \approx 1.53 \times 10^{-4} \text{ s} \tag{14}$$

$$c_{\text{inf,sup}} \approx 1.75 \times 10^{-4} \text{ s} \tag{15}$$

with setup costs

$$C_{\text{train,pinn}} \approx 191.93 \text{ s} \tag{16}$$

$$C_{\text{labels}} + C_{\text{train,sup}} \approx 120.43 \text{ s} \tag{17}$$

These imply approximate break-even thresholds

$$N_{\text{PINN}} \approx 58{,}195 \qquad N_{\text{SUP}} \approx 36{,}765 \tag{18}$$

Table 8: Compressed runtime and deployment-economics summary.

Quantity	Value	Interpretation
FDM representative latency	0.00345 s	Recurring per-query solve cost
PINN inference latency	1.53×10^{-4} s	Faster inference after training
Supervised inference latency	1.75×10^{-4} s	Slightly slower than PINN at inference
PINN setup cost	191.93 s	Offline training investment
Supervised setup cost	120.43 s	Label generation plus training
PINN break-even	58,195 queries	Amortization threshold vs. FDM
Supervised break-even	36,765 queries	Earlier break-even than PINN under current host
Batch throughput at 2048	PINN: 1.89×10^6/s	Exceeds cached FDM under large batches
Batch throughput at 2048	SUP: 1.38×10^6/s	Also exceeds cached FDM
Cached FDM throughput at 2048	5.33×10^5/s	Strong classical implementation remains competitive at small batches

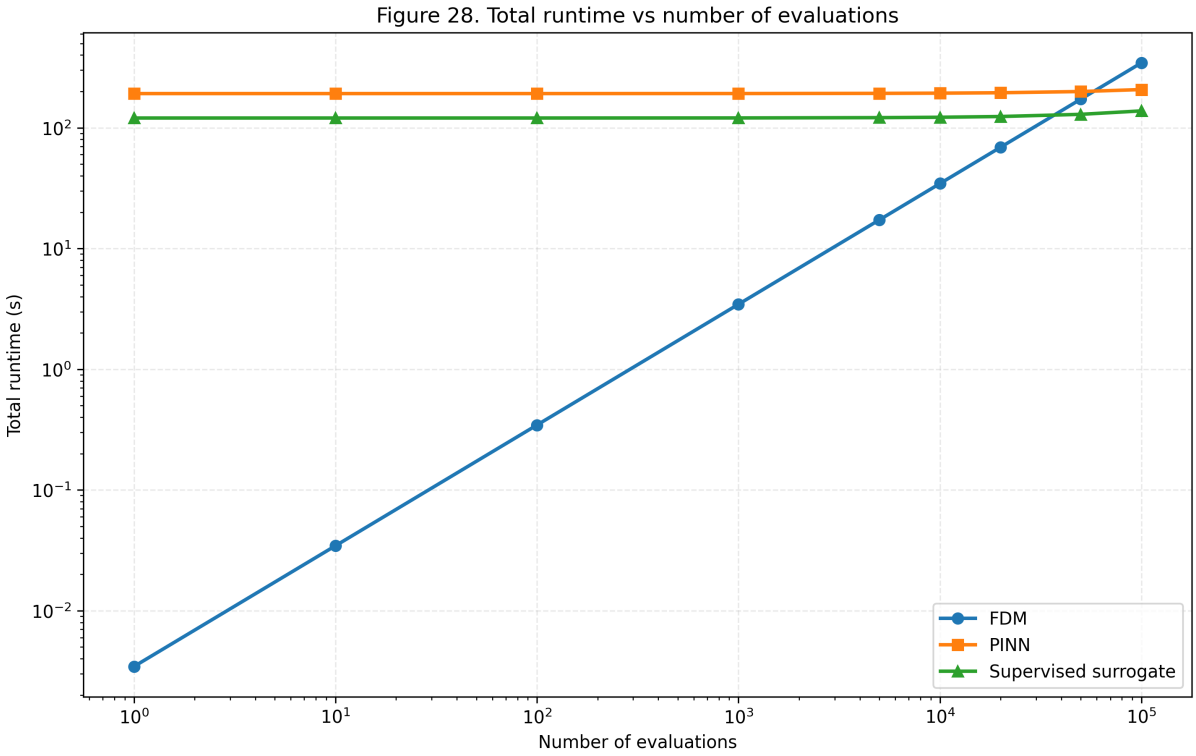

Figure 12: Total runtime versus number of evaluations. Learned surrogates incur setup cost but much smaller slope at large workloads.

Figure 12 shows the core deployment message. At low N, the learned surrogates are dominated by setup cost. At sufficiently large N, the smaller inference slope becomes decisive. Throughput results reinforce this point: at batch size 2048, the PINN reaches about 1.89×10^6 contracts/s and the supervised surrogate about 1.38×10^6 contracts/s, versus about 5.33×10^5 contracts/s for the cached FDM path.

Runtime does not settle solver choice by itself. The supervised surrogate reaches break-even earlier than the PINN under the current host, but it remains much weaker in barrier-sensitive diagnostics. Amortization matters only after admissibility has been considered.

The runtime results should also be read at two levels. The first is the generic workload model summarized by the break-even thresholds. The second is the concrete shape of real workloads, where batching, caching, and repeated evaluation over shared grids can shift the effective crossover point. This is why the paper treats break-even analysis as a decision aid rather than as a substitute for workload-specific profiling.

Table 9: Illustrative workload interpretation under the retained evidence.

Workload type	Main constraint	Preferred current solver
Single-query or audit-style pricing	Local accuracy and direct credibility	FDM benchmark
Large repeated barrier-sensitive grid	Structure and near-barrier Gamma still matter	Barrier-aware PINN
Very large repeated smooth-region workload	Throughput matters more than strict local control	Structured supervised surrogate

Three stylized use cases help interpret this table. For a one-off validation or audit-style pricing task, the benchmark remains preferred because there is little opportunity to amortize training cost and the highest local accuracy is required. For a large repeated barrier risk-surface task, the answer becomes conditional: if the grid is heavily concentrated near the barrier or Greeks matter, the PINN becomes the most defensible learned option. For large repeated smooth-region workloads such as broad screening or approximate scenario generation, the supervised surrogate may become the most economically attractive because it reaches break-even earlier and batches well.

6.2. Solver-Selection Implications

The results lead to a structured decision rule rather than a universal winner. If the task is audit-grade, strongly Greek-sensitive, or low-volume, the FDM benchmark remains preferred. If repeated-query volume becomes very large and the task still cares about barrier structure or near-barrier curvature, the barrier-aware PINN is the most defensible learned model under current evidence. If the workload is very large and tolerated error is materially wider, the supervised surrogate becomes economically attractive.

In short, accuracy determines admissibility and amortization determines economic preference. That is the main operational lesson of the study.

6.3. Limitations and Final Conclusion

The present evidence remains one-dimensional. The test bed is already challenging and diagnostically rich, but it does not imply automatic generalization to higher-dimensional

dynamics, stochastic volatility, jumps, double barriers, or richer path-dependent structures. The residual diagnostic is also intentionally conservative near the barrier; it remains useful as a warning signal, but not yet as a final certification layer.

At the same time, the broader literature already suggests several natural extension paths. Finance-oriented deep-learning methods are being developed for stochastic-volatility, jump-diffusion, and rough-volatility pricing problems, including recent work on Heston-type PINNs, large-scale option surrogates, and neural minimizing-movement methods (Chen et al., 2026; Georgoulis et al., 2026; Hainaut & Casas, 2024; Papapantoleon & Rou, 2025). The present paper should therefore be read as a barrier-focused foundation rather than as a claim to have closed the wider neural-pricing problem.

Within those limits, the conclusions are clear. A strong implicit FDM benchmark remains the standard for local precision in barrier pricing. Naive PINNs fail, and they fail for understandable structural reasons. A barrier-aware PINN with transformed coordinates, a hard-constrained ansatz, and BAAC materially improves learned behavior and becomes the strongest learned model in near-barrier Gamma control and boundary consistency. Supervised surrogates remain attractive for sufficiently large repeated-query workloads, but only under conditional error tolerance. The correct framing is therefore not neural replacement, but disciplined solver selection under joint numerical and economic criteria.

Appendix A. Supplementary Notes

The full thesis version contains the complete analytical formulas, Greek conventions, FDM derivations, hyperparameter tables, residual-estimator definitions, runtime measurement protocol, and reproducibility map. The compressed manuscript keeps only the minimum elements needed to preserve the evidentiary chain.

The analytical reference is the Reiner–Rubinstein down-and-out call solution. Greeks are evaluated numerically relative to analytical truth. The retained acceptance thresholds are

$$\text{Price q95} \leq 2.00\% \tag{A.1}$$

$$\text{Near-barrier Gamma q95} \leq 0.150 \tag{A.2}$$

$$\max |V(B_d, t)| \leq 10^{-5} \tag{A.3}$$

$$\text{Residual q95} \leq 10^{-2} \tag{A.4}$$

$$\text{Positivity violation rate} = 0 \tag{A.5}$$

These criteria are intentionally strict because the objective is not to identify a merely interesting surrogate, but a potentially admissible one. The main implementation facts retained from the long version are:

1. the FDM benchmark uses transformed coordinates, exact barrier alignment, and Rannacher-smoothed Crank–Nicolson;

2. the retained PINN checkpoint comes from the formal ablation workflow rather than from an ad hoc reporting run;

3. the supervised and differential surrogates use hard barrier structure as well, but their gains remain concentrated in smoother regions;

4. runtime results are host- and implementation-specific and should be interpreted as workflow measurements rather than universal constants.

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